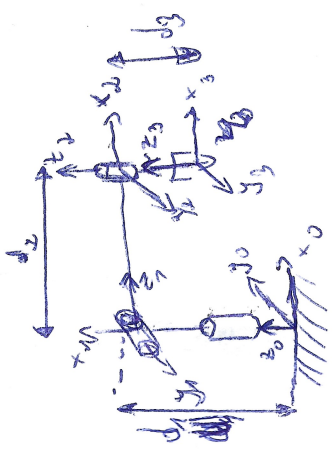


KINEMATIKA PROSTA



KINEMATIKA ODVROTNA

$$x = c_1 (-d_2 s_2 + d_3 c_2) \Rightarrow x = c_1 v$$

$$y = c_1 (-d_2 s_2 + d_3 c_2) \Rightarrow y = c_1 v$$

$$z = d_1 + d_2 c_2 + d_3 s_2 \Rightarrow h = z - d_1$$

$$c_2 = \frac{d_3 v + d_2 h}{d_2^2 + d_3^2}$$

$$s_2 = \frac{-d_2 v + d_3 h}{d_2^2 + d_3^2}$$

$$q_2 = \arctan(s_2, c_2)$$

$$p_2 = \left(\frac{-d_2 v + d_3 h}{d_2^2 + d_3^2}, \frac{d_3 v + d_2 h}{d_2^2 + d_3^2} \right)$$

$$q_3 = 0$$

$$v^2 + h^2 = d_2^2 + d_3^2$$

$$x^2 + y^2 + (z - d_1)^2 = d_2^2 + d_3^2$$

Wartość
osi q potrzebna
początku

$$q = \arctan2(x, y)$$

$$v = \sqrt{x^2 + y^2}$$

$$v = (-d_2 s_2 + d_3 c_2)$$

$$h = d_2 c_2 + d_3 s_2$$

$$A_1 = \begin{bmatrix} c_1 & 0 & s_1 & 0 \\ s_1 & 0 & -c_1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$A_2 = \begin{bmatrix} -c_2 & -s_2 & -c_2 c_3 & -d_2 c_2 c_3 \\ s_2 & c_2 & -s_2 c_3 & -d_2 s_2 c_3 \\ 0 & 0 & c_2 s_3 + d_2 c_3 & 1 \\ 0 & 0 & s_2 s_3 + d_2 s_3 & 0 \end{bmatrix}$$

$$A_3 = \begin{bmatrix} -c_2 s_3 - s_2 c_3 & c_2 c_3 & c_1 (d_2 s_2 + d_3 c_2) \\ -s_2 s_3 + c_2 c_3 & s_2 c_3 & s_1 (-d_2 s_2 + d_3 c_2) \\ c_2 s_3 & -c_2 c_3 & d_1 (d_2 c_2 + d_3 s_2) \\ 0 & 0 & 1 \end{bmatrix}$$



